

Project Design Phase - Solution Architecture

Date: 18 June 2026

Team ID: 24EG105H29

Project Name: AuraCare – Smart Healthcare & Telemedicine Management System

Maximum Marks: 4 Marks

Solution Architecture

AuraCare follows a MERN Stack architecture consisting of React.js frontend, Node.js and Express.js backend, MongoDB database, and JWT-based authentication. The architecture enables secure communication between patients, doctors, hospitals, and administrators through REST APIs.

Architecture Components

Frontend Layer (React.js)

Provides user interfaces for patients, doctors, hospitals, appointment booking, telemedicine, prescriptions, and emergency services.

Backend Layer (Node.js & Express.js)

Handles business logic, authentication, appointment management, prescription generation, and API processing.

Database Layer (MongoDB)

Stores users, patients, doctors, hospitals, appointments, prescriptions, and healthcare records.

Authentication Layer (JWT)

Provides secure login, registration, role-based authorization, and session management.

Telemedicine Module

Supports video consultations, live chat communication, and prescription generation.

Emergency Healthcare Module

Manages ambulance requests, tracking, and emergency assistance services.

Data Flow

Users interact with the React.js frontend. Requests are sent through REST APIs to the Express.js server. The server validates authentication, processes business logic, accesses MongoDB for data storage and retrieval, and returns responses to the frontend for display.